# Section 2.4: Multiplication of Matrices

## Finite Mathematics MTH 132 Spring 2014

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# Scalar Product of a Matrix

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$A =$
$$\begin{bmatrix} 1 & 0 & -3 \\ 4 & -2 & 1 \end{bmatrix}$$
$-5A =$

# Scalar Product of a Matrix

$$A = \begin{bmatrix} 1 & 0 & -3 \\ 4 & -2 & 1 \end{bmatrix}$$

$$-5A =$$

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$0B =$$

# Scalar Product of a Matrix

$A = \begin{bmatrix} 1 & 0 & -3 \\ 4 & -2 & 1 \end{bmatrix}$

$-5A =$

$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$0B =$

$C = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 0 \end{bmatrix}$

$-10C =$

# Matrix Multiplication

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$$\begin{bmatrix} 2 & 3 & -1 \\ 4 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 8 \\ 6 \end{bmatrix}$$

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**When can we find \$AB\$?**

When $A$ is a $m \times n$ and $B$ is a $n \times k$ matrix.

# Matrix Multiplication

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**When can we find $AB$?**

When $A$ is a $m \times n$ and $B$ is a $n \times k$ matrix.

**Find $A$ times $B$**

If the $i^{\text{th}}$ row of $A$ is $\{a_1, a_2, \ldots, a_n\}$ and the $j^{\text{th}}$ column of $B$ is $\{b_1, b_2, \ldots, b_n\}$ then the $i \times j$ entry of $AB$ is: $a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$.

Example 2.4.1.

Find the product

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## Example 2.4.2.

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Example 2.4.3.

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$$\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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### Example 2.4.4.

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$$\begin{bmatrix} 2 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

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